

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 06

Exponents and Logarithms

1 Concepts | 66 synchronized questions

B5 Landscape | one question per teaching page

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Chapter 06 | Exponents and Logarithms

CLASSROOM ROUTE

Concept 1 | Laws of Exponents and Roots**66 questions**

Official lessons: 6-1, 6-2, 6-3, 6-4

The Formula Bank changes automatically at each Concept boundary. Every question ID and order remains synchronized with the approved A4 Classified and Solutions pair.

Concept 1 | Laws of Exponents and Roots

OFFICIAL LESSONS 6-1, 6-2, 6-3, 6-4

Laws of Exponents and Roots

Equation-first recall | use the same rail beside every question in this Concept

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q001 | WRITTEN

C06-K01-6-1-WU1

Find: (a) 10^0 , (b) 2^{-3} , (c) $(0.2)^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V02

Q002 | WRITTEN

C06-K01-6-1-WU2

Let $a \neq 0$, $b \neq 0$. Simplify: (a) $a^{-5}a^8$, (b) $a^5 \div a^{-3}$, (c) $(a^{-1}b)^2$, (d) $(2^2)^3 \div 2^{-2} \times 2^{-3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q003 | WRITTEN

C06-K01-6-1-TRY1

Find: (a) 5^0 , (b) 3^{-2} , (c) $(0.25)^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V03

Q004 | WRITTEN

C06-K01-6-1-TRY2

Simplify the eight source expressions: a^5a^4 , $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, and $2^4 \div (2^{-1})^3 \times 2^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V04

Q005 | WRITTEN

C06-K01-6-1-EX1

Find the nine values: 3^0 , 11^0 , $(2/3)^0$, 4^{-3} , 5^{-3} , 6^{-2} , $(1/3)^{-3}$, $(3/5)^{-2}$, $(0.04)^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V05

Q006 | WRITTEN

C06-K01-6-1-EX2

Simplify the twelve same-base expressions in source Question 2: a^2a^4 , $a^{-2}a^5$, $a^{-3}a^{-4}$, $a^3 \div a^5$, $a^8 \div a^{-3}$, $a^{-6} \div a^{-2}$, $(a^{-1})^3$, $(a^3b^4)^{-2}$, $(a^{-3}b^2)^{-3}$, 3^53^{-3} , 4^44^{-3} , 5^55^{-5} .

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V06

Q007 | WRITTEN

C06-K01-6-1-EX3

Simplify source Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^23^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^42^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, and $(a/2)^3a^{-4} \div (2a)^{-2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V07

Q008 | WRITTEN

C06-K01-6-1-CH

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2} (4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q009 | WRITTEN

C06-K01-6-2-WU1

Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V02

Q010 | WRITTEN

C06-K01-6-2-WU2

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[6]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q011 | WRITTEN

C06-K01-6-2-TRY1

Find $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V03

Q012 | WRITTEN

C06-K01-6-2-TRY2

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, and $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V04

Q013 | WRITTEN

C06-K01-6-2-EX1

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V05

Q014 | WRITTEN

C06-K01-6-2-EX2

Perform the fifteen source calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V06

Q015 | WRITTEN

C06-K01-6-2-CH

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q016 | WRITTEN

C06-K01-6-3-WU1

Express in the form a^r : $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V02

Q017 | WRITTEN

C06-K01-6-3-WU2

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V03

Q018 | WRITTEN

C06-K01-6-3-TRY1

Express in the form a^r : $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V04

Q019 | WRITTEN

C06-K01-6-3-TRY2

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, and $[(27/125)^{1/2}]^{-4/3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V05

Q020 | WRITTEN

C06-K01-6-3-EX1

Express the nine source expressions in the form a^r , including roots of a and numerical bases 2, 3, 5.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V06

Q021 | WRITTEN

C06-K01-6-3-EX2

Evaluate the twelve source expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/25)^{2/5}]^{-5/4}$, $[(81/100)^{-3/4}]^{2/3}$, $[(64/27)^{-4/3}]^{-1/2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V07

Q022 | WRITTEN

C06-K01-6-3-CH

Simplify $\left[\left((1000/27)^{-2/3} \right)^{7/15} \right]^{-15/14}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V08

Q023 | WRITTEN

C06-K01-6-4-WU

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[6]{32} \times \sqrt[3]{16}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V09

Q024 | WRITTEN

C06-K01-6-4-TRY1

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7^3}\sqrt[6]{7^6}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V10

Q025 | WRITTEN

C06-K01-6-4-TRY2

For $a > 0$, $b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V11

Q026 | WRITTEN

C06-K01-6-4-EX1

Perform the twelve numerical calculations in source Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V12

Q027 | WRITTEN

C06-K01-6-4-EX2

For $a > 0$, $b > 0$, simplify the six source expressions built from $\sqrt{a}$, $\sqrt[3]{a}$, $\sqrt[4]{a}$, $\sqrt[6]{a}$ and mixed a, b radicals.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F04 | V01

Q028 | WRITTEN

C06-K01-6-4-CH

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}} \right)^{-3/2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F04 | V02

Q029 | WRITTEN

C06-K01-6-4-REF

Assess the statement $\sqrt{a^2} = a$ for all real a , using positive and negative examples.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q030 | MCQ

C06-K01-6-1-GM01

The value of 2^{-3} is:

A 8

B $1/8$

C -8

D $1/6$

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q031 | MCQ

C06-K01-6-1-GM02

When multiplying equal bases, we:

A subtract exponents

B add exponents

C multiply bases only

D take reciprocals

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q032 | MCQ

C06-K01-6-1-GM03

The expression $a^5 \div a^{-3}$ simplifies to:

A a^2

B a^8

C a^{-8}

D a^{15}

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q033 | MCQ

C06-K01-6-1-GM04

A negative exponent means:

A multiply by -1

B take the reciprocal power

C set the base to zero

D add one

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q034 | MCQ

C06-K01-6-1-GM05

The value of $(0.04)^{-2}$ is:

A 25

B 400

C 625

D 1600

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q035 | MCQ

C06-K01-6-1-GM06

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:

A -2

B -1

C 2

D 6

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q036 | MCQ

C06-K01-6-1-GM07

The positive-exponent form of the 6-1 challenge is:

A a^7b^6

B $1/(a^7b^6)$

C $a^{-7}b^{-6}$

D $1/(a^6b^7)$

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Q037 | MCQ

C06-K01-6-1-GM08

The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:A $1/3$

B 1

C 3

D 9

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q038 | MCQ

C06-K01-6-2-GM01

 $\sqrt[3]{64}$ equals:

A 2

B 4

C 8

D 16

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q039 | MCQ

C06-K01-6-2-GM02

 $\sqrt[3]{12}\sqrt[3]{18}$ equals:

A 4

B 5

C 6

D 9

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q040 | MCQ

C06-K01-6-2-GM03

The fourth root of 1/16 is:

A $1/2$ B $1/4$ C 2 D 4

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q041 | MCQ

C06-K01-6-2-GM04

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:

A $\sqrt[3]{2}$

B $2\sqrt[3]{2}$

C $3\sqrt[3]{2}$

D 0

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q042 | MCQ

C06-K01-6-2-GM05

 $\sqrt[4]{625}$ equals:

A 4

B 5

C 5^2

D 25

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q043 | MCQ

C06-K01-6-2-GM06

When combining roots with the same index, multiply:

A the indices

B the radicands

C the exponents only

D the signs only

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Q044 | MCQ

C06-K01-6-2-GM07

The roots challenge evaluates to:

A 0

B 1

C 2

D 4

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q045 | MCQ

C06-K01-6-3-GM01

 $\sqrt[5]{2^4}$ in rational-index form is:

A $2^{4/5}$

B $2^{5/4}$

C 2^9

D $4^{1/5}$

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q046 | MCQ

C06-K01-6-3-GM02

The value of $27^{2/3}$ is:

A 3

B 6

C 9

D 18

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q047 | MCQ

C06-K01-6-3-GM03

A reciprocal square root is written as:

A $a^{1/2}$

B $a^{-1/2}$

C a^{-2}

D a^2

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q048 | MCQ

C06-K01-6-3-GM04

The value of $81^{-3/4}$ is:

A 27

B $1/9$ C $1/27$ D $1/81$

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q049 | MCQ

C06-K01-6-3-GM05

The exponent of a in $\sqrt[4]{a^3}$ is:A $3/4$ B $4/3$ C $-3/4$

D 3

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q050 | MCQ

C06-K01-6-3-GM06

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:

A 4/9

B 9/4

C 16/9

D 27/64

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q051 | MCQ

C06-K01-6-3-GM07

The 6-3 challenge simplifies to:

A $3/10$ B $10/3$ C $100/9$ D $9/100$

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q052 | MCQ

C06-K01-6-4-GM01

 $2^{2/3} 2^{1/2} \div 2^{7/6}$ equals:A $1/2$

B 1

C 2

D 4

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q053 | MCQ

C06-K01-6-4-GM02

 $4^{5/6} 2^{4/3}$ equals:

A 4

B 6

C 8

D 16

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q054 | MCQ

C06-K01-6-4-GM03

 $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:

A $a^{1/3}$

B $a^{2/3}$

C $a^{5/3}$

D a

WORKING AREA

FORMULA BANK

Zero / negative

$a^0 = 1 \quad a^{-n} = 1/a^n$

Integer laws

$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$

Roots

$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$

Rational indices

$a^{m/n} = \sqrt[n]{a^m}$

Sign control

$\sqrt{a^2} = |a|$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q055 | MCQ

C06-K01-6-4-GM04

The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:

A 2

B 3

C 6

D 12

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q056 | MCQ

C06-K01-6-4-GM05

The source expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

A integer powers

B combining rational exponents

C logarithms

D trigonometry

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Q057 | MCQ

C06-K01-6-4-GM06

The printed decimal in the 6-4 challenge is:

A 0.04

B 0.004

C 0.4

D 4

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F04 | V01

Q058 | MCQ

C06-K01-6-4-GM07

For real a , $\sqrt{a^2}$ equals:A a alwaysB $-a$ alwaysC $|a|$ D 0

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Quiz M1 | QUIZ MCQ

C06-K01-Q-M01

Which law gives $a^m a^n = a^{m+n}$?:

A same-base multiplication

B power of a power

C reciprocal law

D root product

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Quiz M2 | QUIZ MCQ

C06-K01-Q-M02

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

A 2

B 5

C 10

D 125

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Quiz M3 | QUIZ MCQ

C06-K01-Q-M03

The value of $64^{-3/2}$ is:

A 512

B $1/64$ C $1/512$

D 64

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F04 | V01

Quiz M4 | QUIZ MCQ

C06-K01-Q-M04

For a negative real a , $\sqrt{a^2}$ is:A a B $-a$ C 0 D a^2

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F01 | V01

Quiz W1 | QUIZ WRITTEN

C06-K01-Q-W01

Simplify $a^{-3}a^7 \div a^2$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F02 | V01

Quiz W2 | QUIZ WRITTEN

C06-K01-Q-W02

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F03 | V01

Quiz W3 | QUIZ WRITTEN

C06-K01-Q-W03

Evaluate $125^{-2/3}$.

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

Concept 1 | Laws of Exponents and Roots

F04 | V01

Quiz W4 | QUIZ WRITTEN

C06-K01-Q-W04

Explain why $\sqrt{a^2} = a$ is not true for every real a .

WORKING AREA

FORMULA BANK

Zero / negative

$$a^0 = 1 \quad a^{-n} = 1/a^n$$

Integer laws

$$a^m a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

Roots

$$\sqrt[n]{a^m} = a^{m/n} \quad \sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$$

Rational indices

$$a^{m/n} = \sqrt[n]{a^m}$$

Sign control

$$\sqrt{a^2} = |a|$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 06

Exponents and Logarithms

1 Concepts | 50 synchronized questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

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01120009622

Chapter 06 | Exponents and Logarithms

CLASSROOM ROUTE

Concept 2 | Exponential Functions and Their Applications**50 questions**

Official lessons: 6-5, 6-6, 6-7, 6-8, 6-9

The Formula Bank changes automatically at each Concept boundary. Every question ID and order remains synchronized with the approved A4 Classified and Solutions pair.

Concept 2 | Exponential Functions and Their Applications

OFFICIAL LESSONS 6-5, 6-6, 6-7, 6-8, 6-9

Exponential Functions and Their Applications

Equation-first recall | use the same rail beside every question in this Concept

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q001 | WRITTEN

C06-K02-6-5-WU

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F01 | V02

Q002 | WRITTEN

C06-K02-6-5-TRY

Draw $y = 2^x$ and $y = (1/2)^x$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F01 | V03

Q003 | WRITTEN

C06-K02-6-5-EX

Draw the six source graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F01 | V04

Q004 | WRITTEN

C06-K02-6-5-CH

Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q005 | WRITTEN

C06-K02-6-6-WU

Order the source sets 0.7^{-3} , 0.7^{-1} , 0.7^2 and $5/8$, $5/64$, $6/16$ using inequality signs.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F02 | V02

Q006 | WRITTEN

C06-K02-6-6-TRY

Order the three displayed Try sets: 5^0 , 5^{-4} , $5^{3/2}$; $(1/3)^7$, $(1/3)^2$, $(1/3)^{-5}$; and 3^0 , 3^2 , 3^{-2} .

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F02 | V03

Q007 | WRITTEN

C06-K02-6-6-EX1

Order the source sets with bases 3, 2, $3/2$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

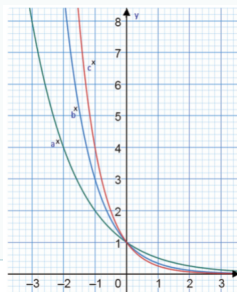
Concept 2 | Exponential Functions and Their Applications

F02 | V04

Q008 | WRITTEN

C06-K02-6-6-CH

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a , b , c .



WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q009 | WRITTEN

C06-K02-6-7-WU

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V02

Q010 | WRITTEN

C06-K02-6-7-TRY

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, and the reciprocal-base inequalities.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V03

Q011 | WRITTEN

C06-K02-6-7-EX

Solve all 24 source equations and inequalities in Exercise (a)–(x), including the 2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125 common-base forms.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V04

Q012 | WRITTEN

C06-K02-6-7-CH

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V05

Q013 | WRITTEN

C06-K02-6-8-WU

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V06

Q014 | WRITTEN

C06-K02-6-8-TRY

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V07

Q015 | WRITTEN

C06-K02-6-8-EX

Solve the twelve source items (a)–(l), each reducible to a quadratic in 2^x or 3^x .

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V08

Q016 | WRITTEN

C06-K02-6-8-CH

Solve completely for real x : $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q017 | WRITTEN

C06-K02-6-9-WU

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x -values.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V02

Q018 | WRITTEN

C06-K02-6-9-TRY

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x -values.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V03

Q019 | WRITTEN

C06-K02-6-9-EX

Solve the three source extrema models: $-3 \leq x \leq 2$, $y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3$, $y = -4 + 2^{2x} - 3^{x+2}$; and the third displayed interval/model.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V04

Q020 | WRITTEN

C06-K02-6-9-CH

For $-1 \leq x \leq 2$, optimise $y = 2 \cdot 4^x - 2^{x+2} + 11$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V05

Q021 | WRITTEN

C06-K02-6-9-REF

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q022 | MCQ

C06-K02-6-5-GM01

Which base gives a strictly decreasing exponential graph?:

A 2

B 3

C $1/2$ D π

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q023 | MCQ

C06-K02-6-5-GM02

The horizontal asymptote of $y = a^x$ is:

A $x = 0$

B $y = 0$

C $y = 1$

D $x = 1$

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q024 | MCQ

C06-K02-6-5-GM03

The range of $y = 5^x$ is:A $\mathbb{R}$ B $[0, \infty)$ C $(0, \infty)$ D $(-\infty, 0)$

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Q025 | MCQ

C06-K02-6-5-GM04

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?:A $(0, \pi)$ B $(1, \pi)$ C $(2, \pi)$ D $(-1, \pi)$

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q026 | MCQ

C06-K02-6-6-GM01

For $0 < a < 1$ and $r < s$, which is true?:

A $a^r < a^s$

B $a^r > a^s$

C $a^r = a^s$

D Cannot compare

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q027 | MCQ

C06-K02-6-6-GM02

The first step when bases differ is to:

A add exponents

B make the bases equal

C take logarithms immediately

D ignore the base

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q028 | MCQ

C06-K02-6-6-GM03

Which is largest? 3^{-2} , 3^0 , 3^2 :A 3^{-2} B 3^0 C 3^2

D They are equal

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Q029 | MCQ

C06-K02-6-6-GM04

The challenge term $(\sqrt[3]{4})^3$ equals:

A $2^{3/2}$

B 2^2

C 2^3

D 4^3

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q030 | MCQ

C06-K02-6-7-GM01

Solving $4^{x-1} = 8^{x+3}$ gives:

A $x = -11$

B $x = -3$

C $x = 3$

D $x = 11$

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q031 | MCQ

C06-K02-6-7-GM02

For $0 < a < 1$, an exponential inequality requires:

A the same exponent order

B a reversed exponent order

C no base conversion

D squaring both sides

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q032 | MCQ

C06-K02-6-7-GM03

The solution of $3^x = 81$ is:

A 2

B 3

C 4

D 5

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q033 | MCQ

C06-K02-6-7-GM04

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

A $0 \leq x \leq 2$

B $x > 2$

C $x < -1$

D $x < 2$

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q034 | MCQ

C06-K02-6-8-GM05

In substitution methods, $t = a^x$ must satisfy:

A $t < 0$

B $t = 0$

C $t > 0$

D $t \in \mathbb{R}$

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q035 | MCQ

C06-K02-6-8-GM06

The positive root of $t^2 - 3t - 16 = 0$ is:

A $(3 - \sqrt{73})/2$

B $(3 + \sqrt{73})/2$

C 8

D 16

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q036 | MCQ

C06-K02-6-8-GM07

The challenge quadratic in t is:

A $t^2 + 5t + 4$

B $t^2 - 5t + 4$

C $t^2 - 4t + 5$

D $t^2 + 4t - 5$

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Q037 | MCQ

C06-K02-6-8-GM08

After solving an exponential quadratic, reject roots with:

A $t > 1$

B $t = 1$

C $t \leq 0$

D $t < 4$

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q038 | MCQ

C06-K02-6-9-GM01

For extrema, the first required range is the range of:

A x B $t = a^x$ C a D y only

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q039 | MCQ

C06-K02-6-9-GM02

The minimum of $y = t^2 - 4t$ at $t = 2$ is:

A -8

B -4

C 0

D 4

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q040 | MCQ

C06-K02-6-9-GM03

The C06-K02 challenge has minimum value:

A 9 at $x = 0$ B 9 at $x = 2$ C 27 at $x = 0$ D 27 at $x = 2$

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q041 | MCQ

C06-K02-6-9-GM04

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:

A 0

B 1

C 2

D It has no minimum

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Q042 | MCQ

C06-K02-6-9-GM05

At which value of x does $a^x + a^{-x}$ attain its minimum?:A $x = -1$ B $x = 0$ C $x = 1$

D It has no stationary point

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Quiz M1 | QUIZ MCQ

C06-K02-Q-M01

Which statement about $y = a^x$ is always true for $a > 0$, $a \neq 1$?:A range $(0, \infty)$

B range all reals

C vertical asymptote

D decreasing for every a

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Quiz M2 | QUIZ MCQ

C06-K02-Q-M02

For base 0.5, increasing the exponent makes the value:

A larger

B smaller

C unchanged

D zero

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Quiz M3 | QUIZ MCQ

C06-K02-Q-M03

The positive-substitution condition is:

A $t < 0$ B $t = 0$ C $t > 0$ D t can be any real

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Quiz M4 | QUIZ MCQ

C06-K02-Q-M04

For $y = 2(t - 1)^2 + 9$, the minimum is:

A 0

B 1

C 9

D 11

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F01 | V01

Quiz W1 | QUIZ WRITTEN

C06-K02-Q-W01

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F02 | V01

Quiz W2 | QUIZ WRITTEN

C06-K02-Q-W02

Order 2^{-3} , 2^0 , $2^{5/2}$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F03 | V01

Quiz W3 | QUIZ WRITTEN

C06-K02-Q-W03

Solve $4^x - 5 \cdot 2^x + 4 = 0$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

Concept 2 | Exponential Functions and Their Applications

F04 | V01

Quiz W4 | QUIZ WRITTEN

C06-K02-Q-W04

Find the minimum of $a^x + a^{-x}$ for $a > 1$.

WORKING AREA

FORMULA BANK

Graph

$$y = a^x \quad a > 0, a \neq 1$$

Growth / decay

$$a > 1 \Rightarrow \nearrow \quad 0 < a < 1 \Rightarrow \searrow$$

Magnitude

$$a > 1 : r < s \Rightarrow a^r < a^s$$

Common base

$$a^u = a^v \Rightarrow u = v \quad t = a^x > 0$$

Extrema

$$t = a^x \quad t\text{-range from } x\text{-domain}$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 06

Exponents and Logarithms

1 Concepts | 74 synchronized questions

B5 Landscape | one question per teaching page

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Chapter 06 | Exponents and Logarithms

CLASSROOM ROUTE

Concept 3 | Introduction to Logarithms and Their Properties**74 questions**

Official lessons: 6-10, 6-11, 6-12, 6-13, 6-14, 6-15, 6-16, 6-17

The Formula Bank changes automatically at each Concept boundary. Every question ID and order remains synchronized with the approved A4 Classified and Solutions pair.

Concept 3 | Introduction to Logarithms and Their Properties

OFFICIAL LESSONS 6-10, 6-11, 6-12, 6-13, 6-14, 6-15, 6-16, 6-17

Introduction to Logarithms and Their Properties

Equation-first recall | use the same rail beside every question in this Concept

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Start with the family cue, write the first move, then use the working grid.

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q001 | WRITTEN

C06-K03-6-10-WU

Convert the source forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V02

Q002 | WRITTEN

C06-K03-6-10-TRY

Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V03

Q003 | WRITTEN

C06-K03-6-10-EX

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V04

Q004 | WRITTEN

C06-K03-6-10-CH

If $\log_5 25 = x$, evaluate $\log_{1/2} x$ without a calculator.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V05

Q005 | WRITTEN

C06-K03-6-11-WU

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1$, $\log_5 5$, $\log_2 8$, $\log_2 16$, $\log_5(1/5)$, $\log_7(49^{3/2})$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V06

Q006 | WRITTEN

C06-K03-6-11-TRY

Find the six Try values $\log_5 1$, $\log_3 3$, $\log_4 4$, $\log_3 27$, $\log_2(1/4)$, $\log_5(1/125)$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V07

Q007 | WRITTEN

C06-K03-6-11-EX

Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V08

Q008 | WRITTEN

C06-K03-6-11-CH

Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q009 | WRITTEN

C06-K03-6-12-WU

Perform $\log_6 2 + \log_6 3$ and $\log_2 30 - \log_2 3 - \log_2 5$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V02

Q010 | WRITTEN

C06-K03-6-12-TRY

Perform the six Try calculations using product, quotient, and power laws, including $4\log_5 3 - 2\log_5 15 - \log_5 45$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V03

Q011 | WRITTEN

C06-K03-6-12-EX

Perform all 18 source expressions (a)+(r), combining same-base logarithms exactly.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V04

Q012 | WRITTEN

C06-K03-6-12-CH

Simplify the fractional expression $\log_5 75 + 3 \log_5 \sqrt{2} - 2 \log_5 \sqrt{6}$ to one reduced fraction.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V05

Q013 | WRITTEN

C06-K03-6-15-WU

Given $\log_{10} 2 = a$ and $\log_{10} 3 = b$, express $\log_{10} 24$, $\log_{10}(27/4)$, $\log_{10} 45$, $\log_{10}(24/75)$ in terms of a , b .

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V06

Q014 | WRITTEN

C06-K03-6-15-TRY

Given $\log_{10} 2 = a$, $\log_{10} 3 = b$, express the four Try arguments 72, $3^8/10$, 15, 12 in terms of a , b .

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V07

Q015 | WRITTEN

C06-K03-6-15-EX

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V08

Q016 | WRITTEN

C06-K03-6-15-CH

If $\log_c 3 = x$, $\log_c 2 = y$, write $\log_c 12$, $\log_c(3/8)$, $\log_c \sqrt{6}$, $\log_2 3$, $\log_{3^b}(2^b c)$ in terms of x , y .

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q017 | WRITTEN

C06-K03-6-13-WU

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V02

Q018 | WRITTEN

C06-K03-6-13-TRY

Evaluate the five Try logarithms using a suitable common base.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V03

Q019 | WRITTEN

C06-K03-6-13-EX

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V04

Q020 | WRITTEN

C06-K03-6-13-CH

Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V05

Q021 | WRITTEN

C06-K03-6-14-WU

Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V06

Q022 | WRITTEN

C06-K03-6-14-TRY

Solve the four Try products of logarithms after making the bases the same.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V07

Q023 | WRITTEN

C06-K03-6-14-EX

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V08

Q024 | WRITTEN

C06-K03-6-14-CH

Simplify $\log_x(\sqrt{y}) \log_{y^2}(z^3) \log_{\sqrt{z}}(x^4)$ for $x > 1$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V09

Q025 | WRITTEN

C06-K03-6-14-REF

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q026 | WRITTEN

C06-K03-6-16-WU

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V02

Q027 | WRITTEN

C06-K03-6-16-TRY

State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and 1/3.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V03

Q028 | WRITTEN

C06-K03-6-16-EX

Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V04

Q029 | WRITTEN

C06-K03-6-16-CH

For $f(x) = \log_5(x - 2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V05

Q030 | WRITTEN

C06-K03-6-17-WU

Order the displayed logarithmic sets, including $\log_2 3$, $\log_2 5$, $2\log_2(1/2)$, $\log_{1/2} 3$, $\log_{1/2} 5$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V06

Q031 | WRITTEN

C06-K03-6-17-TRY

Order the four Try sets of logarithms using monotonicity and change of base.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V07

Q032 | WRITTEN

C06-K03-6-17-EX

Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V08

Q033 | WRITTEN

C06-K03-6-17-CH

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q034 | MCQ

C06-K03-6-10-GM01

 $\log_2 32$ equals:

A 2

B 4

C 5

D 32

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q035 | MCQ

C06-K03-6-10-GM02

 $\log_{1/2} 2$ equals:

A -2

B -1

C 1

D 2

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q036 | MCQ

C06-K03-6-11-GM03

 $\log_5(1/125)$ equals:

A -3

B -2

C 2

D 3

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q037 | MCQ

C06-K03-6-11-GM04

 $\log_7 49^{3/2}$ equals:

A 1

B 2

C 3/2

D 7/2

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q038 | MCQ

C06-K03-6-12-GM01

 $\log_a M + \log_a N$ equals:

A $\log_a(M + N)$

B $\log_a(MN)$

C $\log_a(M/N)$

D $\log_a(M - N)$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$$

Laws

$$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q039 | MCQ

C06-K03-6-12-GM02

The coefficient $3 \log_5 2$ becomes:

A $\log_5(2 + 3)$

B $\log_5 2^3$

C $\log_5(3^2)$

D $3^2 \log_5 2$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q040 | MCQ

C06-K03-6-15-GM03

If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:

A $a + b$

B $2a + 3b$

C $3a + b$

D $3a + 2b$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q041 | MCQ

C06-K03-6-15-GM04

$\log_c \sqrt{6}$ in terms of $x = \log_c 3$, $y = \log_c 2$ is:

A $x + y$

B $(x + y)/2$

C $2x + y$

D $x + 2y$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q042 | MCQ

C06-K03-6-13-GM01

 $\log_{16} 64$ equals:A $1/2$ B $3/2$

C 2

D 4

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q043 | MCQ

C06-K03-6-13-GM02

Change of base gives $\log_a b =$:

A $\log_c a / \log_c b$

B $\log_c b / \log_c a$

C $\log_a b / \log_c b$

D $\log_c a \log_c b$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q044 | MCQ

C06-K03-6-14-GM03

 $\log_4 16 \cdot \log_9 3$ equals:A $1/2$

B 1

C 2

D 4

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q045 | MCQ

C06-K03-6-14-GM04

The challenge product of three logarithms equals:

A 6

B 8

C 12

D 24

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q046 | MCQ

C06-K03-6-16-GM01

The domain of $y = \log_a x$ is:A all real x B $x \geq 0$ C $x > 0$ D $x \neq 0$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q047 | MCQ

C06-K03-6-16-GM02

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

A $x = 0$

B $y = 0$

C $y = x$

D $y = -x$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q048 | MCQ

C06-K03-6-17-GM03

For $a > 1$ and $M < N$, which is true?:

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal

D Cannot compare

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q049 | MCQ

C06-K03-6-17-GM04

For $0 < a < 1$ and $M < N$, which is true?:

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C They are equal

D Both are positive

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q050 | MCQ

C06-K03-6-16-GM05

The vertical asymptote of $y = \log_a x$ is:

A $x = 0$

B $y = 0$

C $x = 1$

D $y = 1$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q051 | MCQ

C06-K03-6-17-GM06

A negative coefficient $-2 \log_a M$ can be written as:

A $\log_a M^{-2}$

B $\log_a(-2M)$

C $\log_a(M - 2)$

D $2 \log_a M$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$$

Laws

$$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q052 | MCQ

C06-K03-6-10-GM05

The base of $\log_a M$ is:A M B a C p D $1/a$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q053 | MCQ

C06-K03-6-10-GM06

The argument of $\log_a M$ is:A a B M C p D 0

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q054 | MCQ

C06-K03-6-11-GM07

 $\log_2(1/8)$ equals:

A -3

B -2

C 2

D 3

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q055 | MCQ

C06-K03-6-11-GM08

 $\log_4 4$ equals:

A 0

B 1

C 2

D 4

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Q056 | MCQ

C06-K03-6-11-GM09

 $\log_3 1$ equals:

A -1

B 0

C 1

D 3

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q057 | MCQ

C06-K03-6-12-GM05

 $\log_a M - \log_a N$ equals:

A $\log_a(MN)$

B $\log_a(M/N)$

C $\log_a(M - N)$

D $\log_a(N/M)$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a(a^p) = p$$

Laws

$$\log_a(MN) = \log_a M + \log_a N \quad \log_a(M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q058 | MCQ

C06-K03-6-12-GM06

 $2 \log_3 5$ equals:

A $\log_3 10$

B $\log_3 25$

C $\log_3 7$

D $\log_3(5/2)$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q059 | MCQ

C06-K03-6-15-GM07

If $a = \log_{10} 2$, $b = \log_{10} 3$, $\log_{10} 12$ is:

A $a + b$

B $2a + b$

C $a + 2b$

D $2a + 2b$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Q060 | MCQ

C06-K03-6-15-GM08

 $\log_{10} 5$ in terms of $a = \log_{10} 2$ is:A a B $1 - a$ C $1 + a$ D $-a$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q061 | MCQ

C06-K03-6-13-GM05

 $\log_8 32$ equals:A $2/3$ B $3/2$ C $5/3$

D 3

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q062 | MCQ

C06-K03-6-13-GM06

 $\log_{25} 125$ equals:A $1/2$ B $3/2$

C 2

D $5/2$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q063 | MCQ

C06-K03-6-14-GM05

 $\log_a b \cdot \log_b a$ equals:

A 0

B 1

C 2

D $\log_a b$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q064 | MCQ

C06-K03-6-14-GM06

 $\log_2 8 \cdot \log_8 2$ equals:A $1/3$

B 1

C 3

D 8

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Q065 | MCQ

C06-K03-6-14-GM07

The best common base for 16 and 64 is:

A 2

B 4

C 8

D 10

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Q066 | MCQ

C06-K03-6-16-GM07

The range of $y = \log_a x$ is:A $x > 0$ B $\mathbb{R}$ C $[0, \infty)$ D $y > 0$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Quiz M1 | QUIZ MCQ

C06-K03-Q-M01

The logarithmic equivalent of $a^p = M$ is:

A $\log_a M = p$

B $\log_p M = a$

C $\log_M a = p$

D $\log_a p = M$

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Quiz M2 | QUIZ MCQ

C06-K03-Q-M02

 $\log_2 8 + \log_2 4$ equals:

A 2

B 3

C 5

D 12

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Quiz M3 | QUIZ MCQ

C06-K03-Q-M03

 $\log_{16} 64$ equals:A $1/2$ B $3/2$

C 2

D 4

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Quiz M4 | QUIZ MCQ

C06-K03-Q-M04

For $a > 1$, $M < N$ implies:

A $\log_a M > \log_a N$

B $\log_a M < \log_a N$

C equality

D no relation

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F01 | V01

Quiz W1 | QUIZ WRITTEN

C06-K03-Q-W01

Find $\log_3(1/27)$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F02 | V01

Quiz W2 | QUIZ WRITTEN

C06-K03-Q-W02

Simplify $\log_5 25 + \log_5 5$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F03 | V01

Quiz W3 | QUIZ WRITTEN

C06-K03-Q-W03

Find $\log_{16} 64$ using change of base.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

Concept 3 | Introduction to Logarithms and Their Properties

F04 | V01

Quiz W4 | QUIZ WRITTEN

C06-K03-Q-W04

State the domain and range of $y = \log_a x$.

WORKING AREA

FORMULA BANK

Definition

$$a^p = M \iff \log_a M = p$$

Values

$$\log_a 1 = 0 \quad \log_a a = 1 \quad \log_a (a^p) = p$$

Laws

$$\log_a (MN) = \log_a M + \log_a N \quad \log_a (M/N) = \log_a M - \log_a N$$

Change base

$$\log_a b = \frac{\log_c b}{\log_c a}$$

Graphs

$$a > 1 : \nearrow \quad 0 < a < 1 : \searrow \quad x = 0$$

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 06

Exponents and Logarithms

1 Concepts | 48 questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

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Chapter 06 | Exponents and Logarithms

CLASSROOM ROUTE

Concept 4 | Applications of Logarithms**48 questions**

Official lessons: 6-18, 6-19, 6-20, 6-21, 6-22

The Formula Bank changes at each Concept boundary.

Concept 4 | Applications of Logarithms

OFFICIAL LESSONS 6-18, 6-19, 6-20, 6-21, 6-22

Applications of Logarithms

Equation-first recall | use the same rail beside every question in this Concept

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Start with the family cue, write the first move, then use the working grid.

Concept 4 | Applications of Logarithms

F01 | V01

Q001 | WRITTEN

C06-K04-6-18-WU

Solve $\log_2 x = 4$.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F01 | V02

Q002 | WRITTEN

C06-K04-6-18-TRY

Solve $\log_2(2x - 1) = \log_2 3$.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F01 | V03

Q003 | WRITTEN

C06-K04-6-18-EX

Solve all 12 source logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F01 | V04

Q004 | WRITTEN

C06-K04-6-18-CH

Solve $\log_3(x+2) - \log_9((x-2)^2) = 1$ completely.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | V01

Q005 | WRITTEN

C06-K04-6-19-WU

Solve $\log_3 x < \log_3 81$ and $\log_{1/2}(x+2) + \log_{1/2}(x-5) > -3$.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | V02

Q006 | WRITTEN

C06-K04-6-19-TRY

Solve the six source logarithmic inequalities using base monotonicity and argument conditions.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | V03

Q007 | WRITTEN

C06-K04-6-19-EX

Solve all 24 source logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | V04

Q008 | WRITTEN

C06-K04-6-19-CH

Solve the source challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | V01

Q009 | WRITTEN

C06-K04-6-20-WU

$$\text{Solve } 5(\log_2 x)^2 - 16\log_2 x + 3 < 0.$$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | V02

Q010 | WRITTEN

C06-K04-6-20-TRY

Solve the three source substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | V03

Q011 | WRITTEN

C06-K04-6-20-EX

Solve the source groups: three equations, three inequalities, and three exact logarithmic values.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | V04

Q012 | WRITTEN

C06-K04-6-20-CH

Solve $(\log_3 x)^2 - \log_3 x - 6 < 0$ for the positive wave-intensity multiplier x .

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | V01

Q013 | WRITTEN

C06-K04-6-21-WU

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | V02

Q014 | WRITTEN

C06-K04-6-21-TRY

Find the maximum and minimum of the source logarithmic quadratic on its stated interval, with the corresponding values of x .

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | V03

Q015 | WRITTEN

C06-K04-6-21-EX

Solve the three source extrema problems, each with a different base and interval.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | V04

Q016 | WRITTEN

C06-K04-6-21-CH

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | V05

Q017 | WRITTEN

C06-K04-6-22-WU

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | V06

Q018 | WRITTEN

C06-K04-6-22-EX

Complete the source digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | V07

Q019 | WRITTEN

C06-K04-6-22-CH

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | V08

Q020 | WRITTEN

C06-K04-6-22-REF

Solve $\log_x(x+2) \leq \log_x(4-x)$ for all real x .

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VMCQ

Q021 | MCQ

C06-K04-6-18-GM01

The argument condition for $\log_3(x - 5)$ is:A $x < 5$ B $x \geq 5$ C $x > 5$ D all real x

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VMCQ

Q022 | MCQ

C06-K04-6-18-GM02

If $\log_2(2x - 1) = \log_2 3$, then $x =$:

A 1

B 2

C 3

D 4

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VMCQ

Q023 | MCQ

C06-K04-6-18-GM03

The valid solution of the challenge equation is:

A 1

B 2

C 4

D 8

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VMCQ

Q024 | MCQ

C06-K04-6-18-GM04

For same-base logarithms, equality of logs gives:

A arguments equal

B bases equal

C arguments add

D no result

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VMCQ

Q025 | MCQ

C06-K04-6-19-GM05

For base $a > 1$, $\log_a M < \log_a N$ means:

A $M > N$ B $M < N$ C $M = N$ D $MN < 0$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VMCQ

Q026 | MCQ

C06-K04-6-19-GM06

For base $0 < a < 1$, the inequality direction is:

A preserved

B reversed

C removed

D squared

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VMCQ

Q027 | MCQ

C06-K04-6-19-GM07

The domain of $\log_{1/2}(x - 5)$ requires:A $x < 5$ B $x \leq 5$ C $x > 5$ D $x \neq 5$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VMCQ

Q028 | MCQ

C06-K04-6-19-GM08

The warm-up intersection is:

A $0 < x < 5$ B $5 < x < 6$ C $x > 6$ D $-3 < x < 6$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VMCQ

Q029 | MCQ

C06-K04-6-20-GM09

The best substitution for a quadratic logarithmic equation is:

A $t = x^2$

B $t = \log_a x$

C $t = a^x$

D $t = 1/x$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VMCQ

Q030 | MCQ

C06-K04-6-20-GM10

The inequality $5t^2 - 16t + 3 < 0$ gives:A $t < 1/5$ B $1/5 < t < 3$ C $t > 3$ D $0 < t < 1$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VMCQ

Q031 | MCQ

C06-K04-6-20-GM11

The substitution variable must satisfy:

A $t > 0$

B $t \in \mathbb{R}$

C $t < 0$

D $t = 1$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VMCQ

Q032 | MCQ

C06-K04-6-20-GM12

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:

A $x < 1/9$ B $1/9 < x < 27$ C $x > 27$ D $0 < x < 3$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q033 | MCQ

C06-K04-6-21-GM13

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

A $t=0$ B $t=1$ C $t=2$ D $t=3$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q034 | MCQ

C06-K04-6-21-GM14

The correct first step for extrema is:

A differentiate in x

B $\text{set } t = \log_a x$

C ignore the interval

D take $x=0$

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q035 | MCQ

C06-K04-6-21-GM15

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:A $[-1,3]$ B $[0,3]$ C $[1,27]$

D all real

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q036 | MCQ

C06-K04-6-21-GM16

The water-tank challenge maximum health rating is:

A 3

B 4

C 5

D 8

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q037 | MCQ

C06-K04-6-22-GM17

The number of digits of N is determined by:

A $\log_{10} N$

B $\lfloor \log_{10} N \rfloor + 1$

C $N+1$

D 10^N

WORKING AREA

FORMULA BANK

Log equations

$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$

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Common log

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Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q038 | MCQ

C06-K04-6-22-GM18

If $-9 < \log_{10} N < -8$, the first non-zero digit is at:

A 8th

B 9th

C 10th

D 11th

WORKING AREA

FORMULA BANK

Log equations

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Log inequalities

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Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q039 | MCQ

C06-K04-6-22-GM19

The common logarithm base is:

A 2

B e

C 10

D any base

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VMCQ

Q040 | MCQ

C06-K04-6-22-GM20

The solution set of $\log_x(x+2) \leq \log_x(4-x)$ is:

A (0,1)

B (1,4)

C (0,4)

D empty

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VQUIZ

Quiz M1 | QUIZ MCQ

C06-K04-Q-M01

Solve $\log_5(x - 1) = 2$:A $x=26$

B 0

C 1

D cannot be determined

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VQUIZ

Quiz M2 | QUIZ MCQ

C06-K04-Q-M02

For $0 < a < 1$ and $M < N$, compare $\log_a M$ and $\log_a N$:

A $\log_a M > \log_a N$

B 0

C 1

D cannot be determined

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VQUIZ

Quiz M3 | QUIZ MCQ

C06-K04-Q-M03

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$:

A $x = 2, \sqrt{2}$

B 0

C 1

D cannot be determined

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VQUIZ

Quiz M4 | QUIZ MCQ

C06-K04-Q-M04

Find the maximum of $y = -(\log_3 x)^2 + 2 \log_3 x + 4$ on $1 \leq x \leq 27$:A 5 at $x=3$

B 0

C 1

D cannot be determined

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F01 | C06-K04-F01-VQUIZ

Quiz W1 | QUIZ WRITTEN

C06-K04-Q-W01

Solve $\log_3(x + 1) = 2$.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F02 | C06-K04-F02-VQUIZ

Quiz W2 | QUIZ WRITTEN

C06-K04-Q-W02

Solve $\log_2 x \geq 3$.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F03 | C06-K04-F03-VQUIZ

Quiz W3 | QUIZ WRITTEN

C06-K04-Q-W03

Solve $(\log_5 x)^2 - 4 < 0$.

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$

Concept 4 | Applications of Logarithms

F04 | C06-K04-F04-VQUIZ

Quiz W4 | QUIZ WRITTEN

C06-K04-Q-W04

How many digits does 6^{20} have using $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$?

WORKING AREA

FORMULA BANK

Log equations

$$\log_a M = \log_a N \Rightarrow M = N \quad (M, N > 0)$$

Log inequalities

$$a > 1 : M < N \quad 0 < a < 1 : M > N$$

Substitution

$$t = \log_a x \quad t \in \mathbb{R} \quad x = a^t$$

Extrema

$$t = \log_a x \quad t\text{-range from the } x\text{-domain}$$

Common log

$$n - 1 < \log_{10} N < n \Rightarrow n \text{ digits}$$